

OptiCrop: Smart Agricultural Production Optimization Engine

PerformanceTesting

Performancetestingevaluateshowyoursystembehavesunderexpectedandpeakloadconditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step1:TestingOverview

Field	Details
TestingToolUsed	Postman
TypeofTesting	Functional/API Testing
TargetModule/API	Crop Recommendation Prediction API
TestEnvironment	Local System(Flask Server)
TestDate	

Step2:TestScenarios

S.No	TestScenario/ Description	No.ofVirtual Users	Duration(sec)	ExpectedOutcome
1	Single user submits crop recommendation request	1	30	Prediction generated successfully
2	Multiple users submit prediction requests simultaneously	20	60	System handles requests without errors
3	Continuous prediction requests under load	50	120	Response time remains acceptable
4	Stress testing with peak user requests	100	180	System remains stable with minimal failures

Step3:PerformanceTestResults

S.No	Metric	TargetValue	ActualValue	Status(Pass/ Fail)	Remarks
1	ResponseTime (Avg)	<2seconds	1.25 seconds	Pass	Fast prediction generation
2	ResponseTime (Max)	<5seconds	3.80 seconds	Pass	Stable under peak load
3	Throughput (Req/sec)		24 req/ sec	Pass	Good request handling capacity
4	ErrorRate	<1%	0.3%	Pass	Very few failed requests
5	CPUUtilization	<80%	65%	Pass	Efficient resource usage
6	MemoryUtilization	<80%	58%	Pass	Memory consumption within limits